BIOMIMETIC LIPID MEMBRANES AS TEMPLATES FOR STRUCTURING INORGANIC SILICA

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Diatoms are single-celled algae, ubiquitous in marine environments, notably they have and exterior cell composed of intricately structured silica (frustule) (**Fig 1**). The control and reproducibility of the hierarchical structuration seen in diatom frustules far exceeds the current abilities of material sciences. Throughout frustule formation, the frustule remains in close contact with lipid membranes, with recent research suggesting that the membrane is an active participant in generating silica structuration.[1] However, lipid structures have remained largely unexplored as biomimetic systems to investigate diatom silica biomineralization. To address this gap, we have developed biomimetic lipid systems in which we use the formation of liquid-ordered (Lo) and liquid-disordered (Ld) domains to direct site-specific silica condensation. We used two model systems to do this; firstly, 2D supported lipid bilayers (SLBs) prepared through Langmuir-Blodgett deposition and studied using TIRF microscopy, we have monitored in real-time silica condensation kinetics, alongside the spatial organization of lipid domains, and the deposited silica. We have evaluated the influence of headgroup composition on silica mineralization. Secondly, 3D giant unilamellar vesicles (GUVs) which have been used to produce structured silica shells. [2,3] We have used fluorescence microscopy alongside NMR spectroscopy to track silica condensation and the chemical interactions between the lipids and silica. Ultimately, this work seeks to provide new insights into the mechanisms governing silica biomineralization and establish a new paradigm for the formation of hierarchically structured materials using phospholipid membranes as templates.

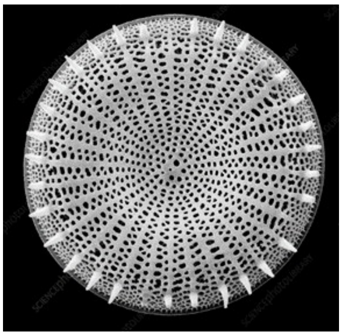

Fig 1. Diatomaceous frustule with structured silica.

REFERENCES

1. Gal *et al*., *Nat. Commun.*, **15**, 7888, (2024), DOI : 10.1038/s41467-024-52211-x
2. Weinberger, *et al*., *Biophysical J*., **105**, 154, (2013), DOI : 10.1016/j.bpj.2013.05.024
3. J Kurniawan, *et al*., *Langmuir*, **34**, 15622, (2018), DOI : 10.1021/acs.langmuir.7b03799